

Homework 3

1. After 200 steps, the average-step velocity stabilizes to around 1.75. The question then asks to modify gamma values to 0.7 and 0.9, but those values are not options in the GUI. I chose 0.667 and 0.919 as close values with noteworthy outcomes. When changing gamma as soon as the robot begins crawling, the robot attains average-step velocity of 0 with gamma = 0.667 and a velocity of 3.33 with gamma = 0.919. A gamma value closer to one allows the DP agent to weigh previous policies more heavily, allowing the agent to value longer-term gain at the cost of increased computation before policy evaluation converges. Visually, the robot takes larger strides with its arms as gamma approaches 1.
2. With an epsilon of 0.739, the robot took around 5 minutes and 20 seconds to cross the screen, intermittently taking strides as it explored a variety of different arm angles at each step. With an epsilon of 0.261, the robot took around 4 minutes and 15 seconds to cross the screen, making small, consistent movements with its arm to nudge it forward. The results align with how epsilon controls the likelihood for the RL agent to choose a less-optimal action and explore.
3. With the default parameters, the reinforcement learning agent took about 3 minutes and 15 seconds at 10 actions per second. On the other hand, an alpha of 0.111 resulted in the agent taking 8 minutes and 20 seconds to cross the screen. A higher learning rate accelerates the update of Q values based on recent observations, which in turn accelerates the robot's movement across the screen.